

# Influence of Paclobutrazol Application on Growth, Flowering, and Vegetative Leaf Anatomy in Potted Chrysanthemum (*Dendranthema grandiflora* cv. Chompoo-Phan)

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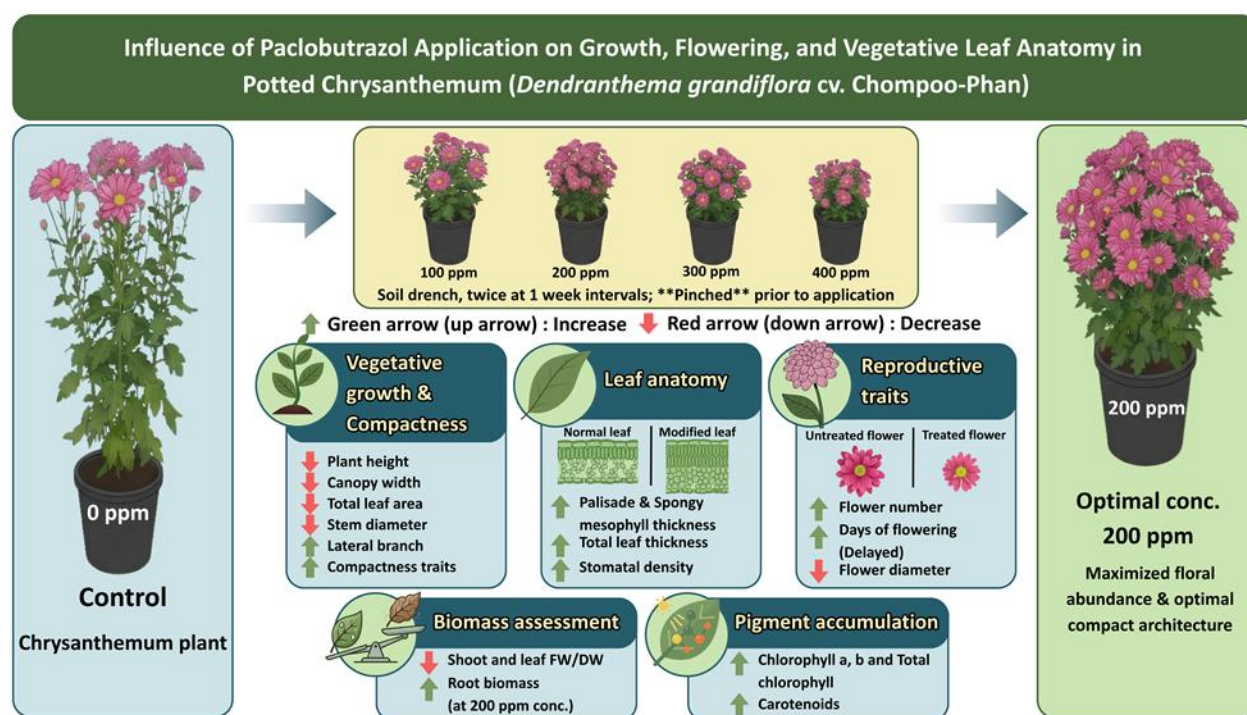
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## GRAPHICAL ABSTRACT



## ABSTRACT

Paclobutrazol (PBZ), a gibberellin biosynthesis inhibitor, was investigated for its effects on vegetative growth, flowering, and leaf anatomy in Chrysanthemum (*Dendranthema grandiflora* cv. Chompoo-Phan). This study evaluated PBZ application at 0, 100, 200, 300, and 400 ppm via soil drenching twice over a 13-week experimental period. Results demonstrated that PBZ significantly decreased plant height, canopy width, leaf area, and stem diameter in a concentration-dependent manner, while promoting lateral branching and enhancing compactness traits. Anatomical assessments revealed distinct leaf modifications, including increased palisade and spongy mesophyll thickness (characterized by increased cell number (hyperplasia) and cell size (hypertrophy)), total leaf thickness, and stomatal density, particularly at higher concentrations. Regarding reproductive traits, flower number increased significantly at 200 ppm, although flower diameter decreased, and flowering time was delayed with increasing PBZ levels. Biomass analysis indicated a physiological trade-off: shoot and leaf fresh and dry weights declined, whereas root biomass increased significantly at moderate concentrations. Furthermore, photosynthetic pigment contents

(chlorophyll *a*, *b*, total chlorophyll, and carotenoids) were enhanced, peaking at 300–400 ppm. Overall, PBZ significantly reduced vegetative growth and delayed flowering while increasing flower number, inducing anatomical reinforcement, and enhancing pigment accumulation. These findings suggest that 200 ppm is the optimal concentration to improve plant compactness and floriferousness, demonstrating high potential for converting this cultivar into a marketable potted plant.

**Keywords:** plant growth retardation, morphology traits, photosynthetic pigment accumulation, leaf anatomy modification, biomass assessment

## 1. INTRODUCTION

Urban expansion in Thailand is rapidly reducing green space, intensifying ecological pressure, and driving demand for nature-based solutions—especially indoor potted ornamental plants for aesthetic, wellness, and air-quality benefits in limited living environments [1]. Globally, the demand for potted plants is also rising due to urbanization, lifestyle changes, and a growing interest in indoor gardening and sustainable living. Although derived from commercial market research—which should be interpreted with caution compared to peer-reviewed or governmental data—recent industry projections estimate this market to grow from USD 1.18 billion in 2025 to USD 1.5 billion by 2030 (accessed on 22 May 2025) [2, 3]. Consumer preferences for potted ornamentals are strongly influenced by traits such as compact form, lush foliage, and floriferousness [4, 5], aligning with the continued growth of Thailand's ornamental plant industry [6, 7].

Chrysanthemum (*Dendranthema grandiflora*) is a globally significant floricultural crop [8], extensively cultivated in Thailand for both the cut-flower and potted-plant markets due to its diverse morphologies, vibrant colors, and extended post-harvest longevity [9, 10]. As a typical short-day plant (SDP), its floral induction is strictly photoperiod-dependent, requiring a continuous dark period that exceeds the critical night length [11, 12]. In Thailand, these highly inductive short-day conditions naturally occur during the winter season, from October to February [9], which is the optimal period for commercial cultivation and high-quality floral development.

Despite this potential, Thailand continues to rely heavily on chrysanthemum imports—valued at approximately USD 9.5 million in 2019—primarily from Malaysia to meet domestic demand. This reliance stems from persistent production challenges: cultivation is geographically restricted to the Northern and Northeastern regions (accounting for only ~10% of the total area), and domestic produce often struggles to match the quality of imported stock [13]. This production gap creates a mismatch with market needs, especially as urbanization drives demand for potted plants. While producing high-value potted chrysanthemums typically involves intensive cultural practices, such as pinching, pruning, or the application of plant growth regulators (PGRs) [14], this study addresses a specific cultivar-related issue. The *in vitro* propagated cultivar used in our research was originally developed for cut-flower production. Consequently, it possesses traits such as tall stems, elongated internodes, and long pedicels that are unsuitable for pot culture, directly conflicting with the compact habit required by the market. In this context, the exogenous application of phytohormones has emerged as an effective approach to improve plant architecture without altering the genetic makeup [15].

Paclobutrazol (PBZ), a plant growth retardant, is a triazole-group inhibitor of gibberellin (GA) biosynthesis that profoundly affects plant growth and development [16]. It functions by inhibiting the oxidation of *ent-kaurene* to *ent-kaurenoic acid* by blocking the cytochrome P450-dependent monooxygenase [17]. While the efficacy of PBZ has been broadly reported across various ornamental crops [18, 19, 20], its application is particularly critical for managing plant architecture within the Asteraceae family. For instance, PBZ has been successfully utilized to control vegetative growth and enhance flowering in related Asteraceae species such as sunflower [21]. Most notably, Chrysanthemum exhibits a highly favorable morphological response to PBZ application. Previous studies demonstrate that PBZ effectively restricts internode elongation and enhances compactness in Chrysanthemum, traits that are essential for high-quality potted plant production [22, 40].

In potted ornamental production, these effects are highly desirable. The primary advantage of PBZ is the effective control of plant height, resulting in a commercially valuable compact morphology [23]. Furthermore, plant photosynthesis is closely correlated with numerous growth indices. Improving photosynthetic activity has been proven to increase the quantity and quality of flowers, flowering longevity, and overall ornamental value. Beyond morphological benefits, PBZ application often induces profound physiological and anatomical enhancements, notably increasing leaf chlorophyll content to create darker green, aesthetically desirable foliage [24]. These external modifications are supported by significant internal anatomical changes; studies on chrysanthemum report that PBZ increases the thickness of palisade and spongy mesophyll layers and total stomatal density. Anatomically, this includes

more numerous, aggregated chloroplasts with distinct grana and increased starch grain accumulation [25]. Collectively, PBZ is an indispensable tool that not only controls plant stature but also enhances overall physiological robustness and aesthetic quality.

Despite the well-documented benefits of PBZ across general chrysanthemum varieties, its specific application on *Dendranthema grandiflora* cv. Chompoo-Phan has never been previously investigated. The 'Chompoo-Phan' is a novel cultivar that was recently developed and registered through chemical-induced mutagenesis combined with *in vitro* tissue culture techniques. Because vastly different morphological responses can be induced by cultivar-specific sensitivities to PBZ, application rates cannot be reliably extrapolated from established varieties to a newly mutated genotype. Therefore, the establishment of precise protocols for 'Chompoo-Phan' is required to ensure its successful morphological adaptation from a tall cut-flower to a compact potted plant.

Furthermore, although general physiological shifts have been described in the existing literature, the anatomical adaptations—such as specific alterations in leaf tissue layers and stomatal architecture—underlying the morphological changes in this newly registered cultivar remain uncharacterized. To address this gap, the efficacy of soil-drenched paclobutrazol at varying concentrations on *Dendranthema grandiflora* cv. Chompoo-Phan was evaluated in this study. Specifically, its effects on vegetative traits, leaf anatomy, flowering phenology, biomass partitioning, and pigment content were comprehensively assessed. Ultimately, the optimal application rate required to transform this newly developed cut-flower cultivar into a high-quality potted plant suitable for the urban market was determined.

## 2. MATERIALS AND METHODS

### 2.1 Plant Materials

In this study, *Dendranthema grandiflora* cv. Chompoo-Phan was selected as the plant material. This novel cultivar was recently developed through chemical-induced mutagenesis combined with *in vitro* propagation techniques at King Mongkut's Institute of Technology Ladkrabang (KMUTL). Furthermore, it has been officially registered and certified by the Department of Agriculture, Ministry of Agriculture and Cooperatives, Thailand. Under standard cultivation, the plant typically exhibits an upright, shrubby cut-flower growth habit, naturally attaining a height of 1.0 to 1.2 m. Because this tall stature is entirely unsuitable for compact potted plant production, morphological modification via the application of plant growth retardants was required for this specific genotype.

### 2.2 Plant Growth Parameters

Morphological parameters were assessed one week after the final application. Plant height was measured from the pot rim to the apical shoot tip, and stem diameter was determined 10 cm above the pot rim using a digital caliper (Model 1108-150, INSIZE). Canopy width was recorded as the widest horizontal foliage span, while the number of primary branches on the main stem was counted. Leaf area was measured using a leaf area meter (LI-3100C, LI-COR, USA), based on twenty randomly selected leaves per replication sampled from the mid-stem region (approximately 15 cm height). The compactness index was calculated as the ratio of mean leaf area divided by plant height (MLA/H), where MLA represents the mean leaf area derived from the 20-leaf sample. During the reproductive stage, data were recorded for the number of flowers per pot, flower diameter (cm), days to first flower bud initiation, and days to 50% flowering, with the latter calculated via linear interpolation.

$$\text{Day to 50\% flowering (DF}_{50}) = \frac{D_1 + (50 - P_1)}{(P_2 - P_1) \times (D_2 - D_1)}$$

Where  $D_1$  = The date of observation just before 50% flowering was reached;  $D_2$  = The date of observation just after 50% flowering was reached;  $P_1$  = The cumulative percentage of flowering observed on date  $D_1$ ;  $P_2$  = The cumulative percentage of flowering observed on date  $D_2$ . These data were recorded from the week of the onset of flower bud emergence until the final week.

### 2.3 Leaf Anatomical Parameters

For anatomical examination, ten leaves were randomly sampled from the middle stem area (~15 cm from the pot rim). Transverse sections were prepared using a freehand sectioning technique. Although this method is non-standardized, section quality and consistency were strictly maintained by ensuring that a single operator prepared all sections throughout the experiment. To prevent selection bias, multiple sections were cut from each sample. Only the thinnest and most intact sections—characterized by a clear, distinct cellular arrangement without any tissue

folding or tearing—were randomly selected for anatomical measurements. These selected sections were immediately mounted unstained in distilled water on glass slides and observed under a compound light microscope (Olympus CX23). The following parameters were quantified: palisade mesophyll thickness, spongy mesophyll thickness, and total leaf thickness. Stomatal examination involved removing the upper epidermis to expose the underlying tissue. The sample was then inverted, and the lower epidermis was observed under a microscope. Chrysanthemum species possess amphistomatic leaves; however, stomatal density is typically greater on the abaxial surface. All measurements were performed using EPview™ software (Version V3.7.7, Olympus). Stomatal density (SD) was calculated as the number of stomata per square millimeter (mm<sup>2</sup>). Stomatal density (SD) was calculated as the number of stomata per square millimeter using the conversion 1 mm<sup>2</sup> = 10<sup>6</sup> μm<sup>2</sup>, following the SI unit definition [26] and methods used by Franks & Beerling [27] and Sack & Buckley [28].

$$\text{Stomatal density per mm}^2 = (N \times 10^6)/A$$

Where N = Number of stomata counted in the microscope image; A = Actual area of the image used for counting.

## 2.4 Determinations of Chlorophyll and Carotenoid Contents

Chlorophyll and carotenoid contents were quantified using a visible spectrophotometer (Model SP380 Plus, Metertech). The investigation of chlorophyll and carotenoids was conducted using a modified approach based on Lichtenthaler's protocol [29]. Ten leaves were randomly sampled, and leaf discs (0.6 cm diameter) were punched and weighed. The leaf tissue was thoroughly homogenized in 10 ml of 80% (v/v) acetone and transferred to a centrifuge tube. The mixture was incubated in the dark for 3 hours. Following incubation, the extract was filtered through Whatman No. 93 filter paper. The final volume was adjusted to 10 ml with 80% acetone, and absorbance was measured at 663, 647, and 470 nm. Pigment concentrations were calculated using standard equations.

$$\text{Chlorophyll } a \text{ (}\mu\text{g/ml)} = 12.25 \times (A_{663}) - 2.79 (A_{647})$$

$$\text{Chlorophyll } b \text{ (}\mu\text{g/ml)} = 21.50 \times (A_{647}) - 5.10 (A_{663})$$

$$\text{Total carotenoids (}\mu\text{g/ml)} = \frac{1000 (A_{470}) - 1.82 (\text{Chlorophyll } a) - 85.02 (\text{Chlorophyll } b)}{198}$$

These concentrations were then used to determine the pigment content per unit leaf area (μg/cm<sup>2</sup>), Total leaf area (A<sub>total</sub>) calculation: The total area was based on 10 discs, each with a radius (r) of 0.3 cm (derived from a 0.6 cm diameter punch).

$$A_{\text{total}} = 10 \times (\pi r^2)$$

$$A_{\text{total}} = 10 \times (3.14 \times (0.3)^2) = 2.826 \text{ cm}^2$$

Content per area calculation: The pigment concentration (μg/ml) was multiplied by the total extract volume (10 ml) to find the total pigment amount (μg/cm<sup>2</sup>), which was then divided by the total leaf area (2.826 cm<sup>2</sup>).

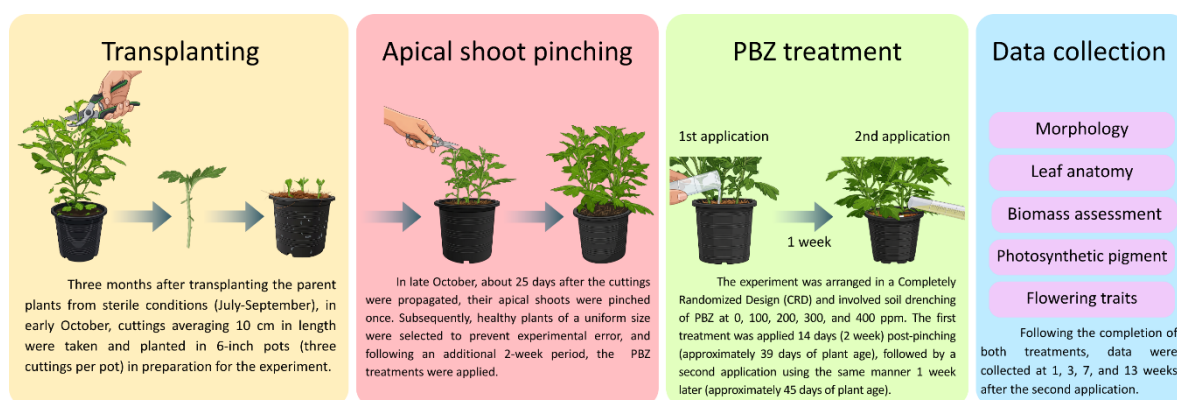
$$\text{Total pigment content (}\mu\text{g/cm}^2\text{)} = \frac{\text{Chlorophyll conc. (}\mu\text{g/ml)} \times 10 \text{ ml}}{2.826 \text{ cm}^2}$$

## 2.5 Biomass Assessment

In the final week, plants were harvested and partitioned into stems, leaves, and roots. Roots were carefully cleaned to remove growth substrates. Fresh weight (FW) of each component was recorded using a precision digital balance (Model BSA3202S-CW, Sartorius). Subsequently, samples were oven-dried at 75°C for 72 hours using a hot air oven (Model FD260, Binder) to determine dry weight (DW). Biomass partitioning was assessed by calculating the shoot-to-root ratio (S/R Ratio).

## 2.6 Experimental Preparation

The proliferation of mother plants commenced in plant tissue culture in April 2024, utilizing a hormone-free MS medium supplemented with 8 g/L agar and 3% sucrose [30, 31]. Upon reaching 1–2 months of age (May to June 2024), the regenerated plantlets were removed from the sterile environment and acclimatized in closed containers to maintain high humidity. Approximately one month later (July 2024), the acclimatized plants were transplanted into 8-inch pots containing a growing medium of chopped coconut husk and coir dust (1:1 ratio) and cultivated in an evaporative-cooling greenhouse. During greenhouse cultivation, plants were maintained using a drip irrigation system that delivered the KMITL2 fertilizer formulation (prepared at a 200× concentration in a 20 L volume). The nutrient solution consisted of Stock A (3.8 kg  $\text{CaNO}_3$  and 80 g Fe-EDDHA) and Stock B (1.8 kg  $\text{KNO}_3$ , 0.65 kg  $\text{KH}_2\text{PO}_4$ , 1 kg  $\text{MgSO}_4$ , and 100 g Nicspay). Irrigation was automatically scheduled four times daily for 3 minutes per session at a flow rate of 30 ml/min, corresponding to a standard 2 L/h emitter output. After four weeks of initial growth, the apical shoots were pinched to promote lateral branching. Following an additional cultivation period (August to September 2024), the stock plants produced mature and uniform branches. In early October 2024, during the natural short-day season in Thailand (which requires no artificial photoperiod induction), 10-cm cuttings were collected from the stock plants and planted in 6-inch pots at a density of 3 cuttings per pot. A comprehensive, chronological timeline of subsequent experimental events is shown in Figure 1. Approximately 25 days after planting (late October 2024), the apical shoots of the cuttings were pinched to ensure uniform growth. For the treatments, a commercial 15% paclobutrazol formulation was diluted to prepare solutions at 100, 200, 300, and 400 ppm, with distilled water as the control (0 ppm). These treatments were applied via soil drenching at 50 ml per pot, twice: the first application at 14 days post-pinching, followed by a second application one week later. The experiment was arranged in a completely randomized design (CRD) with five treatments, three replications, and five pots per replication ( $n=15$ ).



**Figure 1.** Chronological timeline of the experimental procedures for *Dendranthema grandiflora* cv. Chompoo-Phan. The schematic illustrates the key phases conducted under natural short-day conditions, including the initial transplanting of cuttings (early October), apical shoot pinching to promote branching (late October), and the precise schedule of paclobutrazol (PBZ) soil-drench applications (0, 100, 200, 300, and 400 ppm). Data collection for morphological and anatomical assessments was systematically performed at 1, 3, 7, and 13 weeks following the second PBZ treatment.

## 2.7 Data Collection

Data collection was divided into three primary assessments. The first assessment focused on morphological characteristics, including plant height, canopy width, number of branches, the plant compactness index and stem diameter, with data recorded at 1, 7, and 13 weeks after the second treatment (WAT). The second assessment comprised the evaluation of leaf area and pigment content, which were analyzed at 3, 7, and 13 WAT. The third assessment was conducted at 13 WAT during the flowering stage, during which biomass accumulation and overall morphological modifications were systematically recorded.

## 2.8 Data Analysis

Data were subjected to analysis of variance (ANOVA) using SAS 9.0 statistical software (SAS Institute, Inc., Cary, NC.) Mean values were compared using Duncan's Multiple Range Test (DMRT) at the 0.05 significance level.

### 3. RESULTS

#### 3.1 Morphological Traits and Growth Parameters

Chrysanthemum plant height was significantly reduced following the application of paclobutrazol (PBZ), with inhibitory effects detected as early as the first week after treatment. By 7 weeks after treatment (WAT), a height reduction of approximately 73% was observed in plants treated with 100 ppm (20.33 cm) and 200 ppm (21.44 cm). The most substantial reduction—up to 76%—was recorded in the groups treated with 300 ppm (18.35 cm) and 400 ppm (17.92 cm) (Table 1). However, at 13 WAT, a slight increase in plant height was noted across all PBZ-treated groups compared to 7 WAT; this growth was likely associated with floral development during the reproductive phase. Furthermore, no significant differences in plant height were found among the 100–400 ppm treatments at 13 WAT, indicating that the application of PBZ at any of these concentrations was sufficient to suppress stem elongation. Nevertheless, the overall height of the PBZ-treated plants remained significantly lower than that of the untreated control group throughout the experimental period.

In addition to the reduction in plant height, canopy width was significantly influenced by PBZ application. As indicated in Table 1, canopy width was markedly reduced, particularly at 7 (WAT); treatments with 300–400 ppm resulted in an approximately 50% reduction (20.17–20.78 cm) compared to the control (42.00 cm). This suppressive effect was maintained through 13 WAT, where smaller canopy widths were recorded in all PBZ-treated groups (100–400 ppm) compared to the control, although no significant differences were detected among the various PBZ concentrations. Furthermore, regarding branch number (Table 2), significant variations were observed following PBZ treatment. The number of branches increased gradually from 1 WAT to 13 WAT, with treated plants producing approximately 6–7 branches per pot, significantly higher than the 4.5 branches recorded in the control group.

**Table 1.** Effects of PBZ on plant height and canopy width at 1, 7 and 13 WAT in Chrysanthemum.

| PBZ conc.<br>(ppm) | Plant height (cm) |              |             | Canopy width (cm) |              |             |
|--------------------|-------------------|--------------|-------------|-------------------|--------------|-------------|
|                    | 1WAT              | 7WAT         | 13WAT       | 1WAT              | 7WAT         | 13WAT       |
| T1 0 (control)     | 28.31±2.36a       | 74.32±3.47a  | 82.48±3.32a | 27.14±0.98a       | 42.00±1.00a  | 45.00±0.42a |
| T2 100             | 21.13±0.79b       | 20.33±0.75bc | 25.56±0.25b | 23.94±0.38b       | 21.05±0.97bc | 21.42±1.05b |
| T3 200             | 21.84±0.52b       | 21.44±1.55b  | 25.13±1.46b | 22.29±0.16c       | 21.92±0.22b  | 21.14±0.24b |
| T4 300             | 19.27±0.51c       | 18.35±0.80bc | 23.10±0.76b | 21.86±0.79c       | 20.17±0.71c  | 20.97±0.71b |
| T5 400             | 21.98±0.31b       | 17.92±0.43c  | 22.89±0.14b | 21.87±0.73c       | 20.78±0.60bc | 21.01±0.32b |
| F-test             | **                | **           | **          | **                | **           | **          |
| CV %               | 5.43              | 5.40         | 4.30        | 2.67              | 3.00         | 2.38        |

Note: Means ± SD (n = 15), same letters within the column are not significantly different by DMRT. WAT (week after treatment), \*\* indicates highly significant differences at p < 0.01.

**Table 2.** Effects of PBZ on number of primary branches and stem diameter at 1, 7 and 13 WAT in Chrysanthemum.

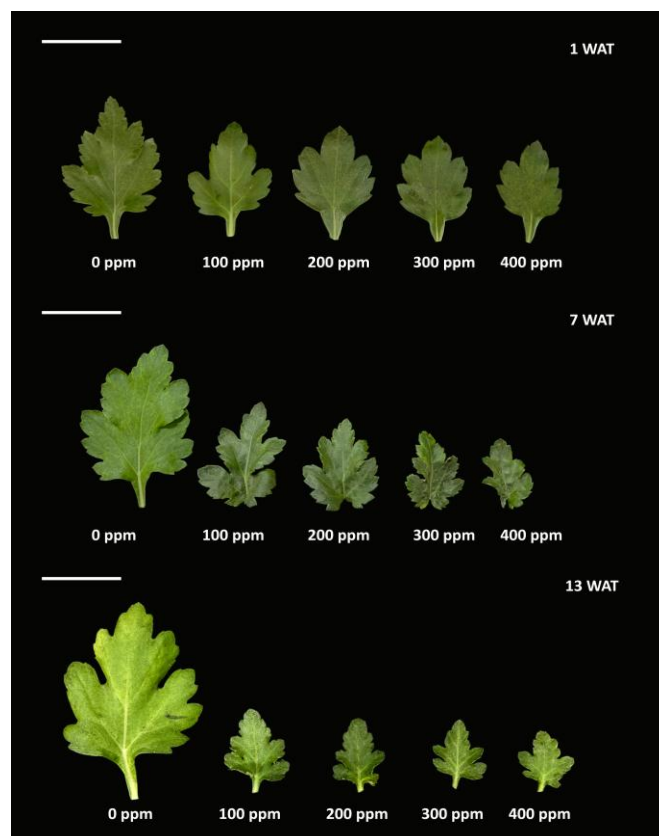
| PBZ conc.<br>(ppm) | Number of branches (branches/pot) |            |            | Stem diameter (mm) |            |             |
|--------------------|-----------------------------------|------------|------------|--------------------|------------|-------------|
|                    | 1WAT                              | 7WAT       | 13WAT      | 1WAT               | 7WAT       | 13WAT       |
| T1 0 (control)     | 3.75±0.77b                        | 4.09±0.95b | 4.49±1.10b | 5.00±0.08          | 5.21±0.42a | 5.76±0.03a  |
| T2 100             | 5.69±0.92a                        | 6.71±0.99a | 7.07±1.08a | 4.59±0.44          | 4.34±0.58b | 4.50±0.29b  |
| T3 200             | 6.33±0.98a                        | 6.96±1.15a | 7.00±1.37a | 4.86±0.20          | 4.13±0.42b | 4.55±0.18b  |
| T4 300             | 5.80±1.11a                        | 6.20±0.84a | 6.95±0.99a | 4.43±0.10          | 4.02±0.48b | 4.04±0.46c  |
| T5 400             | 6.07±0.96a                        | 6.75±0.93a | 6.70±0.97a | 4.55±0.11          | 4.11±1.24b | 4.15±0.07bc |
| F-test             | *                                 | **         | **         | ns                 | **         | *           |
| CV %               | 8.14                              | 6.88       | 9.08       | 5.26               | 5.10       | 5.10        |

Note: Means ± SD (n = 15), same letters within the column are not significantly different by DMRT. WAT (week after treatment), ns indicates no significant difference, \* indicates significant differences at p < 0.05, and \*\* indicates highly significant differences at p < 0.01.

As presented in Table 2, PBZ application significantly reduced stem diameter at 7 and 13 WAT (week after treatment), although no significant difference was observed at 1 WAT. At 7 WAT, stem diameter decreased by approximately 21–23% in plants treated with 300–400 ppm (4.02–4.11 mm) compared to the control (5.21 mm). By 13 WAT, stem diameters in all PBZ-treated groups remained significantly lower than the control. Additionally, unlike

the 7 WAT stage, significant differences were found among the varying PBZ concentrations at 13 WAT, where the 400 ppm treatment resulted in a significantly smaller stem diameter (4.15 mm) compared to the 100–200 ppm treatments (4.50–4.55 mm).

Paclobutrazol (PBZ) has been demonstrated to significantly reduce leaf size and leaf area in various plant species, including members of the Asteraceae family. As presented in Table 3, a significant decrease in total leaf area was observed across all PBZ concentrations by 13 (WAT). This suppressive trend was sustained through 7WAT and 13WAT, with the most pronounced reduction recorded at the highest concentrations (300–400 ppm). At these levels, total leaf area was restricted to 184.35–179.56 cm<sup>2</sup> by 13 WAT, representing a decrease of approximately 80% compared to the control or 920.71 cm<sup>2</sup> (Figure 2).



**Figure 2.** Visual comparison of leaf greenness and morphology of chrysanthemum plants treated with PBZ and the untreated control. Scale bar = 5 cm.

**Table 3.** Effects of PBZ on the mean leaf area of Chrysanthemum.

| PBZ conc. (ppm) | Total leaf area (cm <sup>2</sup> ) |               |               |
|-----------------|------------------------------------|---------------|---------------|
|                 | 3WAT                               | 7WAT          | 13WAT         |
| T1 0 (control)  | 499.01±26.06a                      | 635.27±14.63a | 920.71±73.66a |
| T2 100          | 357.07±18.81b                      | 291.80±14.42b | 266.85±39.63b |
| T3 200          | 352.22±30.83b                      | 292.71±31.56b | 264.44±32.66b |
| T4 300          | 347.29±14.66b                      | 259.35±22.46c | 184.35±11.34c |
| T5 400          | 341.96±12.92b                      | 216.40±19.10d | 179.56±19.94c |
| F-test          | **                                 | **            | **            |
| CV %            | 3.42                               | 1.38          | 5.99          |

Note: Means ± SD (n = 15), same letters within the column are not significantly different by DMRT. WAT (week after treatment), \*\* indicates highly significant differences at p < 0.01.



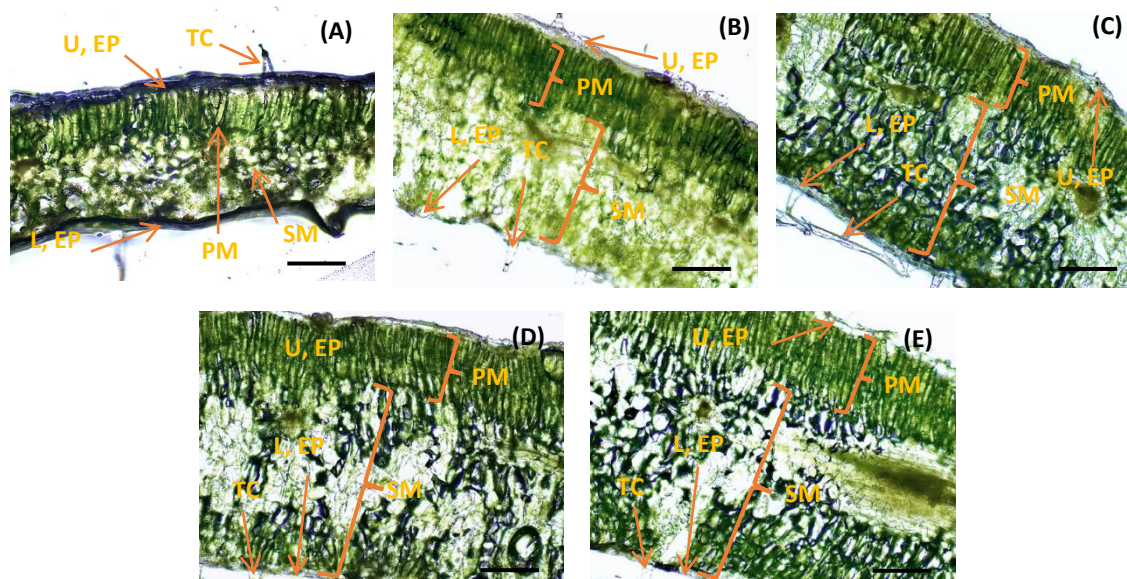
### 3.2 Leaf Anatomy Parameters

As illustrated in Figure 2, a significant reduction in total leaf area was observed in chrysanthemum plants following PBZ treatment. Concurrently, a darker green coloration was noted in the leaves compared to the control samples. To further investigate these morphological alterations, the anatomical characteristics of the leaves were examined. The palisade mesophyll layer was found to be significantly thickened at all PBZ concentrations; specifically, an increase of approximately 83–85% relative to the control was recorded, reaching 101.12–102.85  $\mu\text{m}$  at concentrations of 300–400 ppm (Figure 3). A corresponding enlargement was also observed in the spongy mesophyll layer, where thickness increased dramatically by over 400%. The maximum thickness was detected at 400 ppm (347.26  $\mu\text{m}$ ), representing a value approximately five times greater than that of the untreated plants (Table 4).

**Table 4.** Leaf anatomy and stomatal density of Chrysanthemum plants treated with PBZ in 13WAT.

| PBZ conc. (ppm) | Palisade mesophyll layer ( $\mu\text{m}$ ) | Spongy mesophyll layer ( $\mu\text{m}$ ) | Total leaf Thickness ( $\mu\text{m}$ ) | Stomatal Density ( $\text{mm}^2$ ) |
|-----------------|--------------------------------------------|------------------------------------------|----------------------------------------|------------------------------------|
| T1 0 (control)  | 55.62 $\pm$ 2.98b                          | 67.97 $\pm$ 11.23b                       | 117.08 $\pm$ 7.59b                     | 26.18 $\pm$ 2.52d                  |
| T2 100          | 90.74 $\pm$ 11.89a                         | 297.15 $\pm$ 46.75a                      | 383.54 $\pm$ 56.04a                    | 34.18 $\pm$ 4.29c                  |
| T3 200          | 96.36 $\pm$ 4.91a                          | 306.48 $\pm$ 23.44a                      | 410.18 $\pm$ 29.72a                    | 37.77 $\pm$ 1.73c                  |
| T4 300          | 101.12 $\pm$ 16.57a                        | 309.11 $\pm$ 57.85a                      | 420.02 $\pm$ 44.71a                    | 44.60 $\pm$ 1.01b                  |
| T5 400          | 102.85 $\pm$ 12.44a                        | 347.26 $\pm$ 19.49a                      | 449.43 $\pm$ 7.68a                     | 50.02 $\pm$ 1.09a                  |
| F-test          | **                                         | **                                       | **                                     | **                                 |
| CV %            | 7.80                                       | 13.74                                    | 10.10                                  | 6.35                               |

Note: Means  $\pm$  SD (n = 15), same letters within the column are not significantly different by DMRT. WAT (week after treatment), \*\* indicates highly significant differences at  $p < 0.01$ .

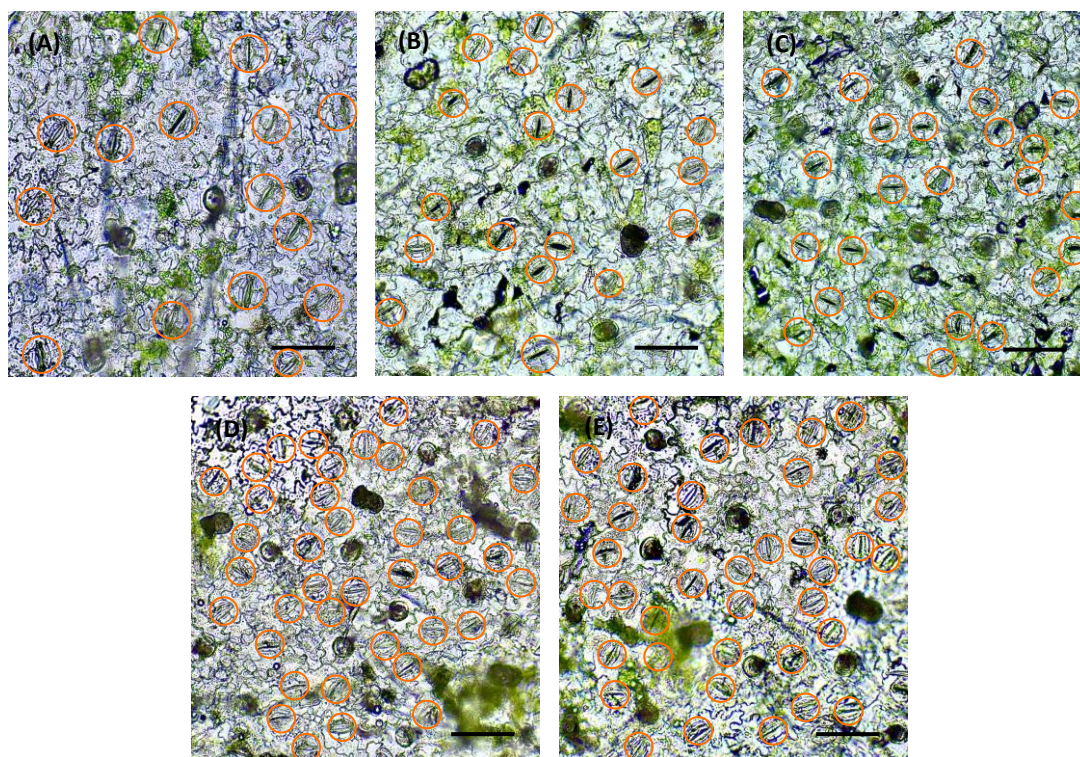


**Figure 3.** Comparison of the leaf anatomy between a control plant and a plant treated with PBZ. Leaf from the control group (A) showing a normal arrangement of cells in the palisade mesophyll (PM) and spongy mesophyll (SM). Leaf from plants treated with PBZ at 100 (B), 200 (C), 300 (D), and 400 (E) ppm, respectively, exhibiting a palisade mesophyll (PM) with cells that are distinctly more elongated and compactly arranged than in the control. U, EP: Upper Epidermis; L, EP: Lower Epidermis; TC: Trichome. Captured at 10x magnification; scale bar = 90  $\mu\text{m}$ .

As indicated in Table 4, a significant increase in total leaf thickness was observed in all PBZ-treated relative to the control. The maximum thickness (449.43  $\mu\text{m}$ ) was recorded in plants treated with 400 ppm PBZ, representing a dramatic increase of approximately 284% compared to the control (117.08  $\mu\text{m}$ ). Regarding stomatal density per unit area ( $\text{mm}^2$ ), a significant increase was found across all PBZ concentrations compared to the control. Through microscopic examination, it was observed that stomatal density increased by approximately 30% at the lowest PBZ



concentration (100 ppm), while a remarkable increase of up to 91% was noted at the highest concentration (400 ppm) (Figure 4). The highest density, approximately 50.02 stomata/mm<sup>2</sup>, was recorded in the 400 ppm treatment compared to 26.18 stomata/mm<sup>2</sup> in the control (Table 4).



**Figure 4.** Light micrographs illustrating stomatal density in leaf epidermal tissues subjected to varying concentrations of PBZ, captured at 10x magnification (scale bar = 90  $\mu$ m). Treatments include: 0 ppm (A), 100 ppm (B), 200 ppm (C), 300 ppm (D), and 400 ppm (E). Progressive alterations in stomatal abundance are evident with increasing PBZ concentration. Orange circles indicate individual stomata highlighted for verification and density quantification.

### 3.3 Biomass Assessment Parameters

It was found that not only stem diameter but also stem fresh weight was significantly reduced by PBZ application across all concentration levels. A reduction of approximately 70% in stem fresh weight was noted at concentrations of 300–400 ppm, where the lowest value was recorded at 42.03 g, compared to 142.73 g in the control group. Similarly, a significant decrease in stem dry weight was observed across all PBZ treatments. Specifically, dry weights at 300 and 400 ppm declined to 9.88 g and 8.61 g, respectively (Table 5). These values represent a reduction of approximately 82% relative to the control, which exhibited the highest dry weight (47.20 g).

A significant and consistent reduction in total leaf fresh weight was also observed across all PBZ concentrations. This reduction was particularly evident at 300 and 400 ppm, where the lowest values were recorded at 69.80 g and 68.27 g, respectively, compared to 87.72 g in the control. A corresponding decrease was noted in total leaf dry weight, which was reduced by 20–36%. The lowest dry weights (9.10 g and 8.91 g) were found in plants treated with 300 and 400 ppm, respectively, relative to 14.00 g in the untreated group (Table 5). Collectively, these findings demonstrate that total leaf biomass in chrysanthemum plants was significantly diminished by the application of PBZ.

**Table 5.** Fresh weight and dry weight in stem, leaves, and roots of Chrysanthemum treated with various paclobutrazol concentrations in 13 WAT.

| PBZ conc. (ppm) | Stem FW (g)   | Total leaf FW (g) | Root FW (g)   | Stem DW (g) | Total leaf DW (g) | Root DW (g) |
|-----------------|---------------|-------------------|---------------|-------------|-------------------|-------------|
| T1 0 (control)  | 142.73±16.03a | 87.72±3.20a       | 29.60±3.23c   | 47.20±5.71a | 14.00±0.73a       | 4.32±0.44c  |
| T2 100          | 59.96±0.86b   | 77.99±3.02ab      | 44.32±13.35ab | 13.08±0.72b | 11.07±1.59b       | 6.81±1.12b  |
| T3 200          | 53.94±1.63bc  | 76.41±0.50ab      | 54.52±0.64a   | 10.78±0.56b | 10.12±0.89bc      | 8.20±1.35a  |
| T4 300          | 43.44±2.48c   | 69.80±1.62b       | 42.38±5.23ab  | 9.88±0.32b  | 9.10±0.90c        | 5.15±0.49c  |
| T5 400          | 42.03±1.05c   | 68.27±3.02b       | 39.57±1.92bc  | 8.61±0.35b  | 8.91±0.04c        | 5.05±0.82c  |
| F-test          | **            | *                 | *             | **          | *                 | **          |
| CV %            | 10.69         | 9.29              | 15.76         | 14.49       | 9.07              | 10.14       |

Note: Means ± SD (n = 15), same letters within the column are not significantly different by DMRT. WAT (week after treatment), \* indicates significant differences at  $p < 0.05$ , and \*\* indicates highly significant differences at  $p < 0.01$ .

Root fresh and dry weights in the examined chrysanthemums were significantly influenced by paclobutrazol (PBZ), exhibiting a distinct concentration-dependent response (Table 5 and Figure 5). Specifically, root fresh weight increased from 29.60 g in the untreated control (0 ppm) to a peak of 54.52 g at 200 ppm PBZ, before declining at higher concentrations (300–400 ppm). A similar trend was noted regarding root dry weight, where values increased from 4.32 g in the control to a maximum of 8.20 g at 200 ppm, before decreasing to approximately 5.00 g at the higher concentrations (Table 5; Figure 5).



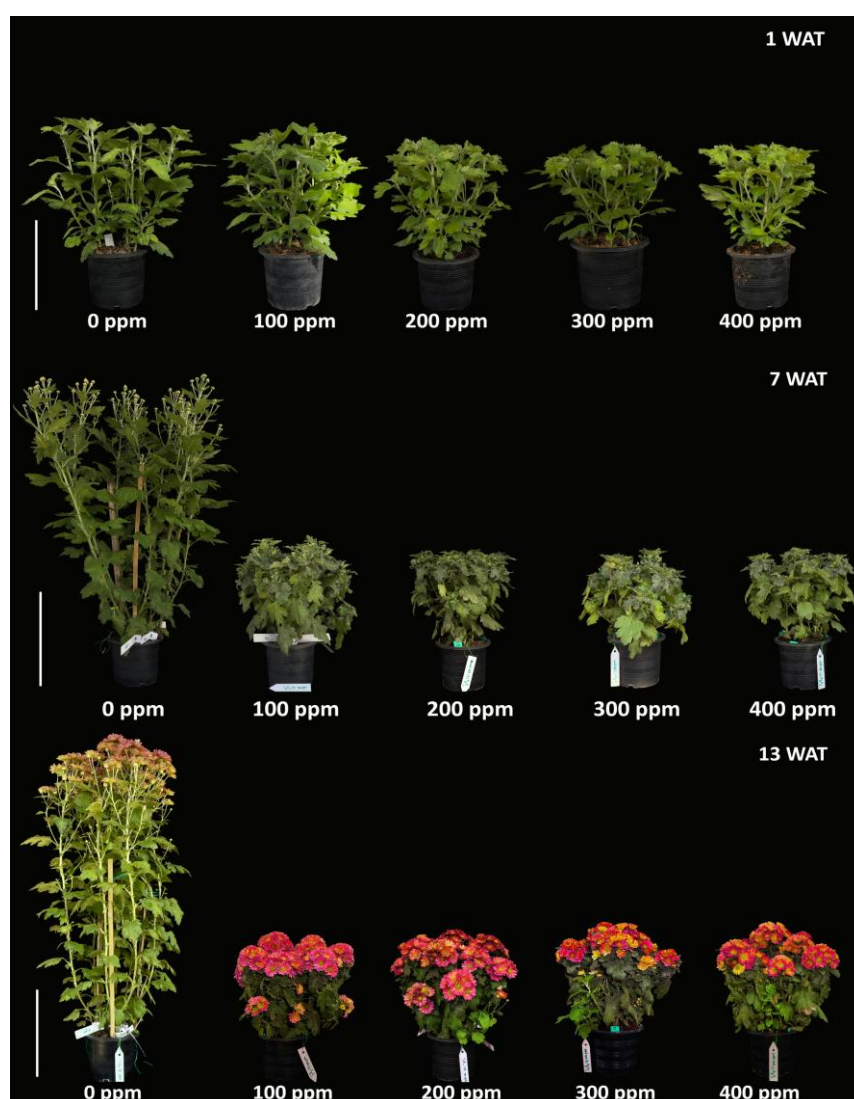
**Figure 5.** Visual comparisons illustrating the effects of varying PBZ concentrations on the overall root architecture of Chrysanthemum. Treatments include: 0 ppm (A), 100 ppm (B), 200 ppm (C), 300 ppm (D), and 400 ppm (E). Notably, PBZ application induced distinct morphological alterations: while the untreated control (A) exhibits thinner, more elongated roots, PBZ treatments resulted in increased root density and root thickening. These enhancement characteristics are most pronounced at 200 ppm (C), which displays a highly branched, compact, and beaded appearance. The scale bars represent 5 cm.



**Table 6.** Periodic evaluation of compactness index and final weeks shoot-to-root ratio of Chrysanthemums treated with PBZ.

| PBZ conc.<br>(ppm) | Shoot to root ratio | Compactness index (cm <sup>2</sup> /cm) |              |             |
|--------------------|---------------------|-----------------------------------------|--------------|-------------|
|                    |                     | 3WAT                                    | 7WAT         | 13WAT       |
| T1 0 (control)     | 13.62±0.45a         | 10.74±1.00c                             | 8.56±0.39d   | 11.18±0.70a |
| T2 100             | 4.49±1.15b          | 15.59±0.26b                             | 11.00±0.39c  | 10.44±0.57a |
| T3 200             | 4.11±1.14b          | 16.61±0.88b                             | 13.45±0.30b  | 10.37±1.05a |
| T4 300             | 3.37±0.34b          | 17.19±0.14a                             | 13.89±0.52ab | 8.06±0.34b  |
| T5 400             | 3.53±0.63b          | 17.90±1.09a                             | 14.36±0.46a  | 7.77±0.50b  |
| F-test             | **                  | **                                      | **           | *           |
| CV %               | 13.76               | 5.01                                    | 3.88         | 7.05        |

Note: Means ± SD (n = 15), same letters within the column are not significantly different by DMRT. WAT (week after treatment), \* indicates significant differences at  $p < 0.05$ , and \*\* indicates highly significant differences at  $p < 0.01$ .



**Figure 6.** Visual comparison of Chrysanthemum morphological responses to varying concentrations of paclobutrazol (PBZ) at 1, 7, and 13 weeks after treatment (WAT). The images are presented at a uniform scale (based on standardized pot dimensions) to accurately illustrate the significant suppression of plant height and canopy width in PBZ-treated plants (100–400 ppm) compared to the untreated control (0 ppm). The scale bar represents 15 cm.

As demonstrated in Table 6 and Figure 5. The application of paclobutrazol to chrysanthemum plants showed a distinct trend in the shoot-to-root ratio across all concentration levels relative to the control group (0 ppm), with a maximum ratio of 13.62. Within the treated plants, this ratio was reduced to a range of 4.49 to 3.37, with the minimum values of 3.37 and 3.53 recorded at 300 and 400 ppm, respectively. Concurrently, by the 3 WAT, the treated plants exhibited a significant increase in the compactness index, from 15.59 to 17.90 cm<sup>2</sup>/cm, compared to the control (10.74 cm<sup>2</sup>/cm) (Table 6). This elevated index was maintained through 7WAT; Despite a marginal decrease across all concentrations, the values remained significantly higher than the control, peaking at 14.36 cm<sup>2</sup>/cm at 400 ppm. However, a significant reversal in this trend occurred by 13 WAT; the control and lower concentration (100 – 200 ppm) maintained the highest compactness index, while the index was lowest for the high concentration (300-400 ppm), which declined sharply to the lowest observed.

### 3.4 Flowering Parameters

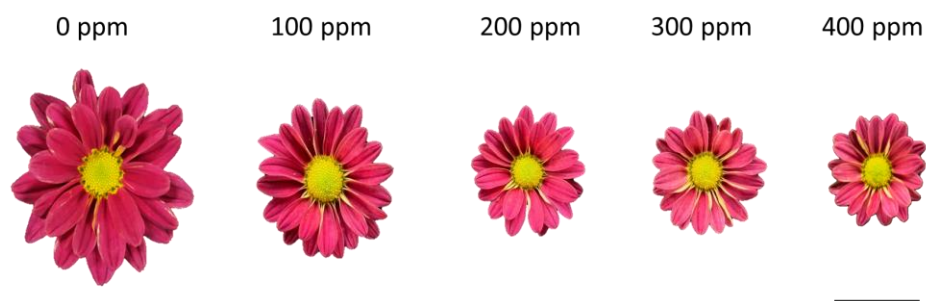
The study of phenological characteristics revealed that the application of PBZ had a statistically significant effect on the number of flowers per plant, flower diameter, days to flower bud initiation, and days to 50% flowering (Table 7). An average of 71.38 flowers/pot and a maximum flower diameter of 4.78 cm were recorded in the control (0 ppm). In contrast, the highest number of flowers (87.51 flowers/pot) was obtained in plants treated with 200 ppm PBZ; this value was significantly different from that of the control. However, a decrease in flower diameter was observed as PBZ concentration increased. The smallest flower size (3.66 cm) was found at 400 ppm (Figure 7), a value significantly smaller than that of the control group. Additionally, the timing of flower bud initiation was found to be delayed with increasing PBZ concentrations. Flower buds developed earliest in the control group (47 days), whereas initiation was delayed to 71 days in the 400-ppm group. Similarly, the time required to reach 50% flowering was significantly extended by high PBZ concentrations. While an average of only 62 days was recorded for the control group, 127 days were required for the 400-ppm group to achieve 50% flowering. In summary, flowering activity was significantly altered by PBZ; specifically, a delayed onset of flower bud initiation and a prolonged period to reach 50% flowering were observed, particularly at concentrations of 300–400 ppm.

**Table 7.** Phenological traits of flowering in Chrysanthemum plants treated with PBZ.

| PBZ conc. (ppm) | Number of Flowers (flower/pot) | Flower Diameter (cm) | First day of flower bud Initiation (day) | Days 50% flowering (day) |
|-----------------|--------------------------------|----------------------|------------------------------------------|--------------------------|
| T1 0 (control)  | 71.38±6.98c                    | 4.78±0.10a           | 47.00±1.00d                              | 61.67±5.69d              |
| T2 100          | 76.46±3.58bc                   | 4.14±0.43b           | 59.33±2.08c                              | 82.33±3.51c              |
| T3 200          | 87.51±5.78a                    | 4.07±0.10b           | 64.67±1.53b                              | 120.30±1.53b             |
| T4 300          | 84.25±3.40ab                   | 3.82±0.08c           | 69.00±2.00a                              | 127.00±2.08a             |
| T5 400          | 81.00±3.83ab                   | 3.66±0.04d           | 71.33±1.53a                              | 127.35±2.00a             |
| F-test          | *                              | **                   | **                                       | **                       |
| CV %            | 6.15                           | 2.04                 | 3.50                                     | 3.20                     |

Note: Means ± SD (n = 15), same letters within the column are not significantly different by DMRT. WAT (week after treatment),

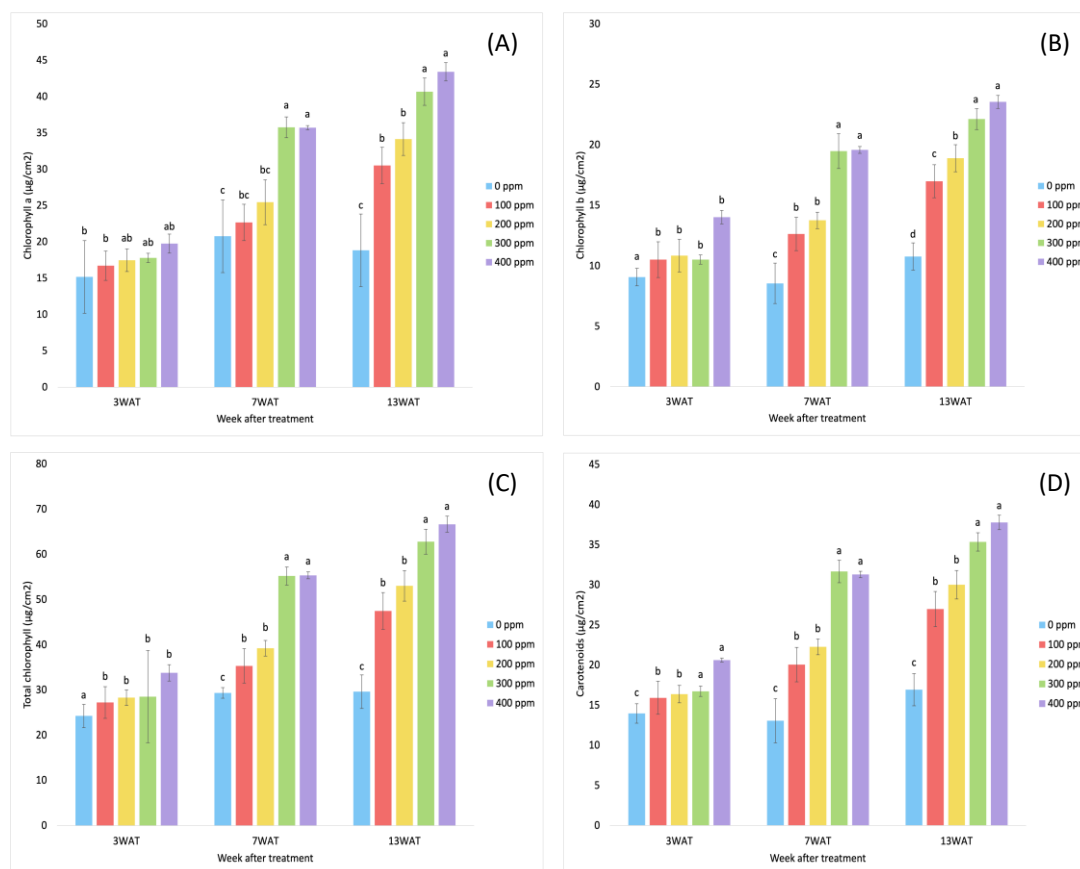
\* indicates significant differences at  $p < 0.05$ , and \*\* indicates highly significant differences at  $p < 0.01$ .



**Figure 7.** Comparison of flower characteristic parameters of chrysanthemums treated with PBZ and the untreated control at the end of the final week. Scale bar = 5 cm.

### 3.5 Photosynthetic Pigment Accumulation Parameters

The results indicated that the content of chlorophyll *a*, chlorophyll *b*, and total chlorophyll was significantly affected by PBZ application, particularly at 7 and 13 (WAT)(Figure 8). Significantly higher pigment contents were recorded in the PBZ-treated groups, especially at 300–400 ppm, compared to the 0 ppm control. The highest total chlorophyll content ( $66.65 \mu\text{g}/\text{cm}^2$ ) was observed at 13 WAT in plants treated with 400 ppm, whereas only  $29.61 \mu\text{g}/\text{cm}^2$  was recorded in the control group (Figure 8C). Regarding chlorophyll *a* and *b*, a continuous increasing trend was noted with higher PBZ levels. This was most evident at 13 WAT under the 400 ppm treatment, where concentrations of  $43.40 \mu\text{g}/\text{cm}^2$  for chlorophyll *a* and  $23.55 \mu\text{g}/\text{cm}^2$  for chlorophyll *b* were obtained (Figure 8A, B). These values represent concentrations nearly double those found in the control group for each parameter.



**Figure 8.** Photosynthetic pigment parameter of chrysanthemum at 3, 7, and 13WAT. Chlorophyll *a* (A), Chlorophyll *b* (B), Total chlorophyll (C), Carotenoids (D); different letters within the sub-figure are significantly different by DMRT ( $p < 0.05$ ). WAT (week after treatment).

Carotenoid content in chrysanthemum leaves was also significantly increased by PBZ application, particularly at 7 and 13 (WAT)(Figure 8D). A continuous rise in carotenoid levels was observed in response to higher PBZ concentrations. The highest accumulation was recorded in the 400-ppm treatment group, reaching  $37.79 \mu\text{g}/\text{cm}^2$  at 13 WAT, compared to  $16.93 \mu\text{g}/\text{cm}^2$  in the control (0 ppm). Similarly, elevated carotenoid levels of  $31.66 \mu\text{g}/\text{cm}^2$  and  $35.36 \mu\text{g}/\text{cm}^2$  were obtained with the 300-ppm treatment at 7 and 13 WAT, respectively; both values were significantly higher than those of the untreated plants. Overall, an enhancement of carotenoid content was clearly demonstrated in leaves treated with increasing concentrations of PBZ, most notably at 300–400 ppm.

### 4. DISCUSSION

The present study demonstrates that paclobutrazol (PBZ) substantially modifies the vegetative, anatomical, physiological, and flowering characteristics of *Dendranthema grandiflora* cv. Chompoo-Phan. The sharp reduction in



plant height, canopy width, leaf area, and stem diameter observed across all PBZ concentrations is consistent with the well-characterized role of PBZ as an inhibitor of gibberellin (GA) biosynthesis, which restricts internode elongation and results in a compact plant stature (Figure 6). These findings corroborate previous reports of height suppression in chrysanthemum [32], marigold [33], zinnia [34], and potted geranium [35].

Notably, a slight increase in plant height was observed in all PBZ-treated groups at 13 (WAT) compared to 7 WAT. This late-stage growth may be associated with floral development during the reproductive phase. Mechanistically, the suppression of GAs by PBZ likely alters the hormonal balance, potentially redistributing cytokinin to support flower stalk elongation via localized cell expansion at the shoot apical meristem [36]. Regarding canopy architecture, the suppressive effect was maintained through 13 WAT; all treated groups (100–400 ppm) exhibited significantly narrower canopies compared to the control, a response consistent with findings in other potted chrysanthemum studies [37, 38].

The reduction in canopy width observed in *Dendranthema grandiflora* cv. Chompoo-Phan is closely associated with the suppression of plant height, primarily driven by the restriction of gibberellin (GA) activity [17]. Upon PBZ application, internode elongation is inhibited due to reduced cell expansion in the sub-apical meristem, resulting in compressed internodes and a more compact plant architecture [39, 22]. This morphological response may be associated with the inhibition of gibberellin (GA) biosynthesis at the transcriptional level. Although specific molecular pathways were not directly evaluated in the present study, in distantly related species such as *Jatropha curcas*, the downregulation of key genes in the GA biosynthetic pathway—specifically *ent-kaurene synthase (KS)*, *ent-kaurene oxidase (KO)*, and *GA20-oxidase (GA20ox1)*—has been demonstrated following PBZ treatment [40]. Because the suppression of these genes is known to limit the production of bioactive GAs, cell division and internode elongation are subsequently inhibited. Consequently, by analogy with these molecular observations, it is postulated that a similar mechanism might operate in chrysanthemum, which could potentially explain the compact, dwarf phenotype and the shortened internodes observed in this study.

Furthermore, the enhanced branching observed across all PBZ treatments reflects a reduction in apical dominance following GA inhibition. This finding is consistent with previous studies on chrysanthemum, which reported increased lateral bud break and outgrowth subsequent to PBZ application [41]. Although branch number increased, a substantial decrease in leaf size and total leaf area was observed, indicating a morphological shift toward smaller but more numerous foliage elements. This pattern is commonly reported in potted chrysanthemums treated with PBZ, where compact clusters of small leaves contribute to a desirable canopy density [42]. Importantly, while the compactness index declined slightly in the final stages (13 WAT) compared to the peak vegetative phase, PBZ treatment effectively maintained a dense, market-preferred pot form compared to the looser architecture of untreated plants.

Microscopic analysis revealed that PBZ application significantly altered the leaf anatomical structure of *Dendranthema grandiflora* cv. Chompoo-Phan, resulting in a marked increase in total leaf thickness and stomatal density compared to the control. The maximum leaf thickness was recorded at 449.43  $\mu\text{m}$  in plants treated with 400 ppm PBZ, corresponding to a substantial expansion of both mesophyll tissues (Table 4). Notably, observations from the cross-sections (Figure 3) indicated a distinct structural modification: the palisade mesophyll transitioned from a single cell layer in control plants to a double layer in PBZ-treated leaves. This hyperplasia, combined with the elongation of palisade cells (hypertrophy), is a characteristic response to triazole growth retardants described in chrysanthemum, where treated plants exhibit thicker leaves with increased cell packing density [39]. Notably, observations of the cross-sections (Figure 3) indicated a distinct structural modification: the palisade mesophyll transitioned from a single cell layer in control plants to a double layer in PBZ-treated leaves. This hyperplasia, combined with the elongation of palisade cells (hypertrophy), is recognized as a characteristic response to triazole growth retardants in chrysanthemum, where thicker leaves with increased cell packing density are typically exhibited [45]. It is hypothesized that these structural modifications represent a physiological adaptation to GA deficiency, potentially mediated by DELLA proteins. Although molecular interactions were not investigated in the current study, by analogy with molecular studies in the distantly related Euphorbiaceae family [40]—where the expression of *RGA-LIKE (RGL)*, a gene encoding a DELLA protein, was upregulated by PBZ treatment—it is postulated that a similar regulatory mechanism might operate in chrysanthemum. Because the accumulation of DELLA proteins is known to act as a negative regulator of GA signaling, cellular elongation is restricted while differentiation and radial expansion are promoted. Consequently, it is theorized that this comparable mechanism could potentially underlie the development of the thicker, robust leaves and the compacted cellular arrangement observed in the treated plants.

Physiologically, this thickening is attributed to the inhibition of gibberellin biosynthesis; the reduction in active GAs suppresses longitudinal cell elongation while promoting anticlinal cell division and radial expansion, leading to

the development of additional mesophyll layers [16, 39]. Furthermore, the treatment significantly enhanced stomatal density, peaking at 50.02 stomata/mm<sup>2</sup> at 400 ppm compared to 26.18 stomata/mm<sup>2</sup> in the control. This increase was likely attributed to the reduction in leaf area, leading to a denser packing of stomatal complexes rather than an absolute increase in stomatal initiation. Mechanistically, as PBZ restricts epidermal cell elongation, the overall leaf surface area is reduced. However, stomatal differentiation persists or decreases at a disproportionately lower rate compared to cell expansion. Consequently, this differential growth results in a mechanical concentration of stomata within the limited leaf area [44]. This phenomenon aligns with previous anatomical observations in chrysanthemum, where epidermal cell expansion was significantly reduced following PBZ treatment, yielding a compacted tissue structure [45]. It is postulated that this anatomical reinforcement—characterized by thicker leaves with increased photosynthetic tissue volume and higher stomatal frequency—functions as a physiological adaptation to optimize gas exchange and water-use efficiency (WUE) within a reduced leaf area. Although physiological parameters such as WUE and photosynthetic performance were not directly quantified in the present study, it is hypothesized that a similar mechanism might operate in this newly developed cultivar. By analogy with physiological enhancements documented in distantly related ornamental species such as *Paeonia lactiflora* [15], the anatomical modifications observed here could potentially facilitate improved photosynthetic capacity and WUE, thereby contributing to the overall quality and stress tolerance of the potted plant.

The comprehensive assessment of biomass accumulation (Table 5) illustrates that PBZ induces a fundamental physiological trade-off in resource allocation, shifting priority from aerial vegetative growth to subterranean development. A distinct, concentration-dependent suppression was observed in the shoot system. Specifically, stem fresh weight declined markedly from 142.73 g in the untreated control to 42.03 g at 400 ppm. A similar trend was noted for stem dry weight, which decreased from 47.20 g to 8.61 g. Furthermore, foliage mass was significantly reduced following PBZ application. Total leaf fresh weight decreased from 87.72 g in the control to 68.27 g at 400 ppm, while leaf dry weight fell correspondingly from 14.00 g to 8.91 g. This simultaneous inhibition of stem and leaf biomass is primarily driven by the blockade of gibberellin (GA) biosynthesis, which restricts cell elongation and diminishes the sink strength of the canopy, thereby limiting carbon investment in vertical expansion [19].

In sharp contrast, the root system exhibited a robust stimulatory response; root fresh weight and dry weight significantly increased to peak at 54.52 g and 8.20 g in the 200-ppm treatment, respectively, compared to the control (29.60 g FW and 4.32 g DW). This inverse relationship is governed by the mechanism of assimilate repartitioning: the suppression of shoot demand allows a greater surplus of photoassimilates to be translocated to the roots [46, 47]. However, at supra-optimal concentrations (>300 ppm), the drastic reduction in total photosynthetic area and excessive inhibition of cell elongation likely constrained overall plant productivity, limiting the resources available for root accumulation.

Concurrent with this biomass shift, a transformation resembling contractile roots—characterized by thickening, shortening, and a unique segmented or beaded appearance—was observed [46]. Anatomically, this altered morphology is generally attributed to the inhibition of GA biosynthesis, where longitudinal cell elongation is restricted while compensatory radial expansion (hypertrophy) and increased transverse cell division (hyperplasia) within the cortical parenchyma are promoted. Although endogenous hormone concentrations were not quantified in the current study, it is hypothesized that these structural changes might be driven by an internal hormonal rebalancing. Based on physiological models established in previous research [16], it is postulated that maintained endogenous auxin levels, alongside increased abscisic acid (ABA) and cytokinins, could potentially promote lateral root initiation and vascular differentiation. Functionally, these thicker, sturdier roots are presumed to serve a dual purpose by enhancing physical anchorage and providing an enlarged storage volume for water and carbohydrates. Furthermore, by analogy with functional adaptations documented in other ornamental species such as *Paeonia lactiflora* [15], it is hypothesized that these specific root modifications might ultimately improve hydraulic conductivity and enhance survival rates under water-deficit conditions.

Furthermore, this physiological reconfiguration was quantitatively evidenced by a significant reduction in the shoot-to-root ratio, which decreased from 11.18 in the control to 7.77–8.06 in treated plants (Table 6). This shift confirms that PBZ does not simply retard growth but actively prioritizes root establishment [48]. Through complex hormonal crosstalk; recent molecular studies indicate that the downregulation of GAs is coordinated with the upregulation of auxin and abscisic acid signaling pathways in the root zone. This establishes a functional equilibrium where a compact, low-transpiration canopy is supported by a robust, highly efficient root system, a trait essential for the stress resilience and post-transplant performance of potted ornamentals [49].

The evaluation of reproductive traits (Table 7) and visual characteristics (Figures 6 and 7) revealed a distinct physiological trade-off: floral abundance increased with PBZ treatment at the expense of individual flower size and

phenological timing. Specifically, the maximum flower number (87.51 flowers/pot) was achieved with the 200-ppm treatment, which was likely driven by the suppression of apical dominance and the subsequent promotion of lateral branching. Furthermore, it is hypothesized that this significant floriferousness might be attributed to an alteration in endogenous hormonal homeostasis, specifically the crosstalk between gibberellins and cytokinins. Although transcriptomic analyses were not conducted in the current study, by analogy with molecular observations in distantly related species within the Euphorbiaceae family [40], it is postulated that a similar regulatory mechanism might operate in chrysanthemum. In those studies, the rapid upregulation of *Cytokinin Oxidase/Dehydrogenase* (*CKX1* and *CKX5*) was induced by PBZ treatment. While *CKX* genes are typically associated with cytokinin degradation, it is theorized that their modulation in response to GA inhibition could represent a complex regulatory feedback loop. In the context of this study, such a comparable mechanism could potentially enhance floral primordia differentiation, thereby explaining the increased floral abundance observed. Thus, the observed increase in floral abundance in Chrysanthemum likely results from PBZ-induced hormonal rebalancing that favors reproductive development over vegetative growth [36]. It concurrently reduced flower diameter from 4.78 cm in the control to 3.66 cm. This reduction in floral organ dimensions is a direct morphological consequence of restricted cell elongation in the petals, a response recently confirmed in potted chrysanthemum, where PBZ significantly constrained inflorescence expansion [42].

Furthermore, despite the observed increase in flower number, the substantial delay in reaching 50% flowering (extending to 127 days at 400 ppm) indicates that PBZ interferes with the floral transition. By inhibiting gibberellin biosynthesis, PBZ appears to retard the differentiation of floral primordia and slow the rate of anthesis, a phenomenon well-documented to extend the production cycle in Asteraceae species [44, 50]. This phenological delay might be associated with PBZ interference with the expression of floral integrator genes. Although gene expression profiles were not analyzed in the present study, by analogy with molecular evidence from distantly related species [40]—where PBZ treatment significantly downregulated the expression of Suppressor of Overexpression of *CO1* (*SOC1*)—it is postulated that a similar mechanism might operate in chrysanthemum. Because *SOC1* functions as a crucial transcription factor that integrates floral signals to promote the transition from the vegetative to the reproductive phase, it is hypothesized that its potential suppression—consequent to reduced GAs levels—could provide a plausible molecular explanation for the prolonged production cycle.

Consequently, while plant architecture and floriferousness are significantly improved by PBZ, the associated delay in flowering requires growers to carefully schedule their operations. To address this, several practical mitigation strategies can be implemented. First, growers should optimize application rates by selecting lower PBZ concentrations, which effectively enhance flower size and yield. Regarding timing, the application schedule can be appropriately adjusted, for instance, by delaying the treatment until 4 weeks after pinching. In cases where excessive PBZ application results in severe stunting, a foliar application of exogenous gibberellic acid (GA3) can serve as a corrective measure to compensate for the localized inhibition of gibberellin biosynthesis in meristematic tissues. Furthermore, growers can proactively adjust their production calendars—such as advancing the planting date or initiating short-day forcing—to offset the expected delay in anthesis and ensure the harvest aligns with market demand.

The analysis of photosynthetic pigments (Figure 7) reveals that Paclobutrazol (PBZ) application significantly enhances the biochemical quality of the foliage, resulting in a characteristic dark green phenotype. All pigment fractions—chlorophyll *a*, chlorophyll *b*, total chlorophyll, and carotenoids—exhibited a concentration-dependent increase, with the highest values consistently recorded at 400 ppm across all measurement periods (3, 7, and 13 WAT). Physiologically, this intensification of greenness is driven by two synergistic mechanisms. First, the concentration effect: the PBZ-induced reduction in leaf area, coupled with the thickening of the palisade mesophyll as observed in our anatomical analysis, leads to a denser packing of chloroplasts per unit leaf area [15]. Second, and more critically, PBZ alters the hormonal regulation of pigment metabolism. By inhibiting gibberellin biosynthesis, PBZ indirectly stimulates the endogenous levels of cytokinins, which are potent promoters of chloroplast differentiation and chlorophyll biosynthesis [51]. Furthermore, cytokinins actively retard chlorophyll degradation (senescence) by stabilizing thylakoid membranes, thereby maintaining high photosynthetic competence even at maturity [52, 53]. The parallel increase in carotenoids (Figure 7D) is equally significant; as accessory pigments and antioxidants, their accumulation suggests an upregulation of the isoprenoid pathway, enhancing the plant's photoprotective capacity against oxidative stress [16, 54, 55].

Although paclobutrazol (PBZ) is globally recognized and extensively utilized as a plant growth retardant across numerous fruit trees and ornamental potted plants, including various chrysanthemum varieties within the Asteraceae family, cv. 'Chompoo-Phan' presents a unique case. Specifically developed for cut-flower production through

chemically induced mutagenesis combined with in vitro tissue culture techniques, the potential of this newly registered cultivar for potted plant production has never been previously evaluated. Because responses to PBZ are highly variable depending on plant species, application methods, and concentrations, the specific efficacy of PBZ on this novel genotype was investigated in this study. A highly positive response to PBZ application was observed in cv. 'Chompoo-Phan'. The influence of PBZ not only significantly reduced plant height and induced a more compact canopy, but also increased the number of flowers, which is a highly desirable characteristic for potted ornamentals.

Additionally, these morphological changes were accompanied by significant anatomical adaptations in the leaves. Collectively, these data demonstrate a significant shift in growth characteristics from a vigorous cut-flower variety to a compact potted ornamental form. Ultimately, this research provides crucial foundational data for the application of plant growth retardants to newly developed, mutagenesis-derived chrysanthemum varieties, offering significant potential for commercialization and future applications.

## 5. CONCLUSIONS

This study provides clear evidence that paclobutrazol (PBZ) effectively restricts vegetative growth while enhancing the anatomical and physiological quality of Chrysanthemum (*Dendranthema grandiflora* cv. Chompoo-Phan). PBZ application significantly reduced plant height and canopy width by inhibiting internode elongation, thereby transforming a standard cut-flower cultivar into a compact potted form. Anatomically, the treatment induced distinct leaf hyperplasia and hypertrophy, characterized by thicker mesophyll layers, increased stomatal density, and elevated chlorophyll content, alongside a vigorous root system capable of supporting the plant under restricted growth conditions.

Balancing the trade-off between vegetative retardation and reproductive performance, 200 ppm is recommended as the optimal concentration for commercial application. This rate maximized branch development, root biomass, and flower number (87.51 flowers/pot) without causing the excessive flowering delay observed at higher concentrations (300–400 ppm). Ultimately, the application of 200 ppm PBZ successfully facilitates the conversion of this traditional cut-flower cultivar into a compact, highly aesthetic, and commercially viable potted plant, directly addressing the growing demand within urban markets.

Future research should further explore the physiological mechanisms underlying these responses, particularly focusing on gas exchange efficiency, hormone profiling, and the interaction between PBZ and photoperiod management to refine year-round production strategies. Additionally, assessments of post-production shelf life and a comprehensive economic analysis comparing PBZ application costs against market value improvements are highly recommended to establish these commercial production protocols fully.

## ACKNOWLEDGMENTS

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## AUTHOR CONTRIBUTIONS

**Kanjana Saetiew:** Conceptualization, Methodology, Resources, Project administration, Supervision, Funding acquisition. **Phatchayamon Bhurivardhanayodhin:** Data curation, Writing - Original draft preparation. **Vasan Sean-in and Nakorn Boonnoi:** resources and investigation. **Montinee Teerarak:** Formal analysis, Writing- Reviewing and Editing.

## CONFLICT OF INTEREST STATEMENT

The authors declare no conflicting interests.

## DECLARATION OF USE OF GENERATIVE AI

During the preparation of this manuscript, the authors used Gemini 3.1 Pro to improve language readability and check grammar. The author carefully reviewed all AI-edited text and takes full responsibility for the final manuscript.

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